

作业

8.10 两等幅馈电的半波振子沿 y 方向排列:

$$(1) \quad d = \lambda/4, \quad \varphi = \pi/2; \quad (2) \quad d = \lambda/4, \quad \varphi = -\pi/2.$$

它们的辐射功率都为 $2W$, 计算上述两种情况在 xy 平面内 $\varphi=30^\circ$ 、 $r=1km$ 处场强值。

8.12 由两同相激励的偶极子天线组成的天线阵, 间距 $d = 1.5\lambda$, 求 $\theta = \pi/2$ 平面内的辐射方向图。

8.13 一线天线阵共有 40 个单元, 每单元相距 $\lambda/2$, 各单元激励电流大小相等且同相, 计算波束宽度及第一个零点位置。

解: $N = 40, \quad d = \lambda/2, \quad kd = \pi, \quad \psi = 0, \quad \omega = Nd = 20\lambda$

$$|F(\varphi)| = \frac{1}{N} \left| \frac{\sin \frac{Nkd \sin \varphi}{2}}{\sin \frac{kd \sin \varphi}{2}} \right| = \frac{1}{40} \left| \frac{\sin(20\pi \sin \varphi)}{\sin \frac{\pi}{2} \sin \varphi} \right|$$

$$\text{因为当 } ws' = \pm 0.443, \quad \text{sinc } ws' = 0.707. \quad (ws')_{B/2} = \frac{w}{\lambda} \sin \frac{\varphi_B}{2} = \frac{Nd}{\lambda} \sin \frac{\varphi_B}{2} = 20 \sin \frac{\varphi_B}{2}$$

所以波束宽度为:

$$\frac{20\lambda}{\lambda} \sin \frac{\varphi_B}{2} = 0.443$$

$$\sin \frac{\varphi_B}{2} = 0.02215$$

$$\frac{\varphi_B}{2} = 1.2692^\circ$$

$$\varphi_B = 2.5384^\circ$$

$$\text{第一个零点位置时 } \frac{20\lambda}{\lambda} \sin \frac{\varphi_0}{2} = 1, \quad \sin \frac{\varphi_0}{2} = 0.05, \quad \frac{\varphi_0}{2} = 2.866^\circ, \quad \varphi_0 = 5.732^\circ$$

8.15 假定由于口面场非均匀分布, 抛物面天线有效面积只是实际面积的 60%, 如要得到 45dB 增益, 计算抛物面天线直径。

(a) 500 MHz; (b) 40GHz.

$$\text{解 } G_D = 10^{\frac{45}{10}} = 10^{4.5} = 3.16 \times 10^4$$

$$(1) \quad f = 500\text{MHz}, \quad A_e = \frac{G_D \lambda^2}{4\pi} = 905.73\text{m}^2$$

$$A' = \frac{A_e}{60\%} = \frac{5}{3} A_e = 1509.55\text{m}^2$$

$$d = \sqrt{\frac{4A'}{\pi}} = 43.9\text{m}$$

$$(2) \quad f = 40\text{GHz}, \quad A_e = \frac{G_D \lambda^2}{4\pi} = 0.142\text{m}^2$$

$$A' = \frac{A_e}{60\%} = 0.236m^2$$

$$d = \sqrt{\frac{4A'}{\pi}} = 0.548m^2$$